



DALHOUSIE UNIVERSITY

Faculty of Engineering

Department of Electrical and Computer Engineering
ECED3901 — Electrical Engineering Design II

Project Definition & Preliminary Design Report

Prepared by

Team Three (3)

Presented to

Dr. Vincent Sieben

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Question 1:

The final deliverable is defined by a specific set of parameters and constraints, set by the client, which are designed to ensure team safety and consistency. Below is a comprehensive breakdown of the rules and constraints that will govern our workflow and decision-making:

- The robot must operate safely and pose no danger.
- The course must not be modified.
- No pneumatic or hydraulic actuation as well as no gas or aerosol systems
- No voltages over 24V
- Only one robot may be deployed
- Each group gets three attempts or a total of 15 minutes on the course
- We may only use parts provided in the kit or parts bought with the team's budget
- The diameter and height of the robot must be less than 550 cm
- The robot must be autonomous and ready by 9am on the challenge day
- We must choose either a coastal zone which is clear of pirates but a longer path or an open ocean zone which is shorter but may encounter trouble along the route.
- The pirates can be pushed to the opposite side by activating any of the 4 panels in the open ocean map.
- There will be a simulated life raft placed randomly that may be rescued
- The team will be randomly selected to be A or B. The robot must be able to run from all positions

Team 3 must perform extensive R&D to deduce how to meet the requirements while following the constraints. The client has provided a practice map which allows the team to best test and innovate our design.

Simulations testing possible implementations of all subsystems will be done for the robot. As an example, there are multiple ways to implement the cargo handling system. Likewise, Team 3 must decide if it would be better to create a design which uses the grooves in the cargo to pick it up, or if using magnets to pick it up by the washers would be more optimal.

The team is provided with electrical components, including a LiDAR sensor, as well as a budget which may use to purchase more components.

Team 3 has several members with programming experience, which will prove valuable for designing the robot. Furthermore, there are various members whose strengths lie in writing. This will be useful for writing reports and documentation. As for the weaknesses which the group must overcome, none of us currently have experience using ROS. However, this is something that the team will learn throughout the project.

Question 2:

Beginning with our stakeholder definition, there are two major groups:

- Group One, being the members of Team 3, producing the deliverable,
- The acceptee of the deliverable, Dr Vincent Sieben.

Each member of Team 3 expects to produce the hardware and software needed to produce a LiDAR driven ROS2-based robot, integrated into an existing chassis and based on an X86 based compute platform. This robot will act as a freight cargo vessel. Team 3 members expect to be challenged, participate in the learning process by creation, and research, and expect teamwork, communication and collaboration from each other.

The client of the final deliverable, Dr Vincent Sieben, expects a functional robot cargo vessel, capable of navigating a series of navigational and logistical challenges, operating in a preset course.

The final robot design shall:

- | | |
|---|---|
| • Include Safety Lighting Modules | ↑ |
| • Have the ability to reach a port, from various pre-set destinations | ↑ |
| • Be able to transport cargo to such a target port | ↑ |
| • Be able to pick up new cargo | ↑ |
| • Rescue overboard personnel | ↑ |
| • Deliver & return cargos, in a timely manner (<3 minutes) | ↑ |
| • Not lose cargo (<i>cumulative, max. × 2</i>) | ↓ |
| • Not engage in any physical contact with the wall, or other “watercraft” | ↓ |

The client, Dr Sieben, expects this deliverable to act as an academic exercise, and for all members of Team 3 to build familiarity with all aspects required for completion. The client expects all members will participate, and equally gain knowledge. There are of course risks involved in the completion of the project, which are listed below:

Team 3:

- Failure of completion of the project. Whether a hardware, management, social, or time failure, it is a possibility that Team 3 will not be able to complete a subsystem, or the entire project, for reasons not currently evident, or not present, in the required allotment of time.
- Work mismanagement. Certain group members having to take on more they can handle, and overburden themselves. This can lead to burnout, mistakes, and decreased performance in other classes.

The Client (Dr Sieben):

- Liability and Safety Risk. Dr Sieben is ultimately responsible for the safety of all involved, and the work/learning. When student groups work end up working with the provided hardware, being soldering irons, lithium batteries, or moving mechanical parts, there will be a number of inherent physical risk that will have to be planned for, and mitigated.
- No Return on Investment. All resources—including lab space, all provided materials and budgets, & the client's own time are finite, and. The client is “investing” these assets into Team 3 with the expectation of a specific end product and educational outcome.

Regarding the specific risk management topic of unsafe navigation leading to damage to Team 3's robot and/or surrounding objects, Team 3 has taken this into consideration, and considered the following risk probabilities. A simple risk probability matrix can be found below, identifying metrics for likelihood and severity:

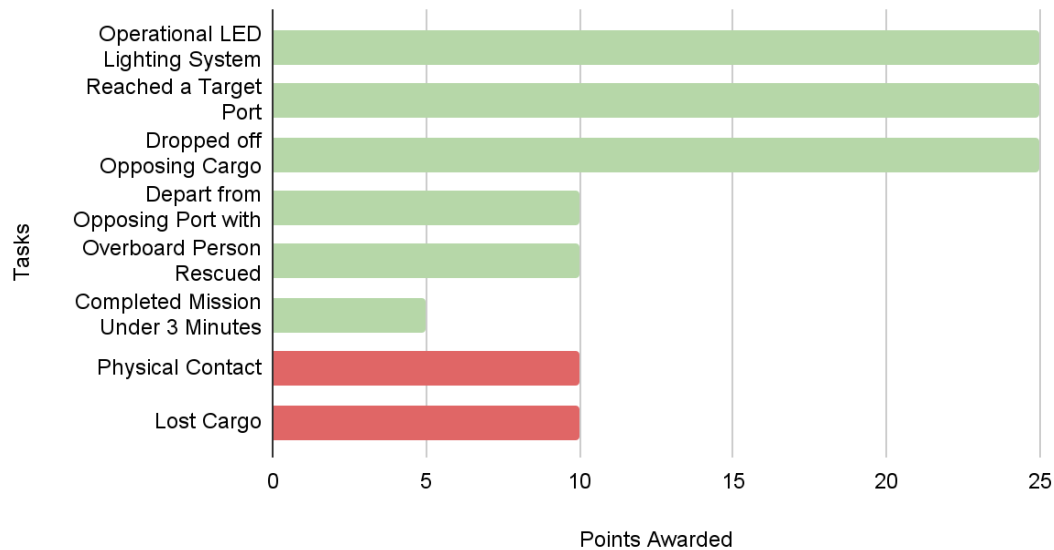
		Severity		
		Critical	Moderate	Marginal
Probability	High	Medium Risk	Low Risk	Low Risk
	Med	Medium Risk	Medium Risk	Low Risk
	Low	High Risk	Medium Risk	Low Risk

Risk Matrix: Probability description of unsafe navigation leading to equipment damage

All low risk items will need to be monitored, medium risk items planned for, and high risk items have a mitigation strategy for.

A Risk Tornado Diagram is a specialized style bar chart, frequently used in sensitivity analysis to visualize how different uncertain variables impact a specific project outcome (*such as a total cost or project finish date*). It is referred to as a “tornado” diagram, as the data categories are listed vertically and ordered by the magnitude of their impact, creating a funnel shape that is widest at the top. Team 3 has a prepared a tornado diagram detailing challenge scoring metrics, and such is presented on the following page:

Half Tornado Diagram - Points Awarded vs Task



Half Tornado Diagram: Risk Management & Design Strategies

Team 3 plans to make heavy use of simulation, primarily through Gazebo and RViz2 LiDAR visualization, to assist in robot pathfinding, driving calibration, and object detection algorithms. Simulations have already been conducted, and all group members are currently using these tools through a mix of Windows Subsystem for Linux, and Distrobox packages. Team 3 has prepared the robot's simulator with the ROS2-GZB bridge, and has achieved LiDAR object visualization through RViz2 using said bridge.

As a current top priority, Team 3 will be measuring and modelling the course 3D mode for simulation use.

A potentially significant risk to the project would be a sizeable discrepancy appearing between the Gazebo simulation and Team 3's working hardware. A program tested in the Gazebo environment may not be interpreted by the robot as expected, compared to the robot in the real space. Therefore, a foresightful test will be to frequently upload the instructions to the physical hardware to observe the actual behaviour of the robot. This will mitigate the accumulation of error between actual and simulated testing environments.

Hardware degradation will occur over the course of the semester as tests accumulate. It is impossible to prevent hardware degradation directly. Instead, mitigation in the form of **A)** balancing real world and simulated test environments to reduce the usage of the physical parts, and **B)** keeping accurate performance measurements of the components used over time to keep track of which components are degrading and by how much.

The motors used for the robot may incur a major problem that will lead to issues in regard to navigation when using dead reckoning. This issue arises if each wheel does not have the same output when supplied with the same amount of energy. The motors will have to be individually tested and have the mismatch accounted for in software, by supplying differing amounts of energy to each motor.

Question 3:

The ROS2 is a software development kit/interface that contains pre-written software libraries and tools to help program complex robot behaviours. The ROS2 ecosystem is a standard for robot application development in industries, as it contains numerous open source tool sets for prototyping — to production of robotic systems.

ROS2 has a variety of features in its framework that will bring forth the best in our design.

- Data Distribution Service

This acts as a communication pipeline interface with nodes.

- Nodes and Topics

Nodes are code files that use ROS driver functionalities, and each communication pipeline are referred to as “topics”. This code can be written in any language, the official libraries support C, C++ and python, but other community supported languages can be used, such as Java. For our proposal, Team 3 will be primarily using C++ as all team members familiar with the functionality.

- Communication Methods

- 1) **Publisher subscriber method:** *(will be elaborated later.)*
- 2) **Service method:** Node A requests Node B for an asset and Node B sends a response.
- 3) **Action method:** Node A sets a goal and Node B replies with feedback until the final act has been completed and then sends the result.

- .bag files
- Parameter Configuration

ROS2 parameters, which are associated with specific nodes that configure driver settings without changing the source code. This will be helpful when using the LiDAR sensor.

- Transformations (tf2)
- Clock

As we are integrating LiDAR, hence we will need to bridge LiDAR's and our system's clock and synchronize them, most likely using SteadyTime. In a publisher and subscriber method of communication between 2 nodes, the publisher node sends out a file/code as a message through the pipeline (DDS), to the receiving end, referred to as a subscriber. Each pipeline is a topic, and the publisher can choose to send messages to multiple subscribers.

Question 4:

ROS2 stands for “Robot Operating System 2”. It is described as a set of software libraries and tools for building robot applications. More specifically, a publisher-subscriber architecture is implemented, where publishers send requests to “topics”, where they are further passed to subscribers. A single topic can handle several publishers and subscribers simultaneously, lending to a robust, easily configured system. The three topics “/cmd_vel”, “/odom” and “/scan” will be discussed henceforth:

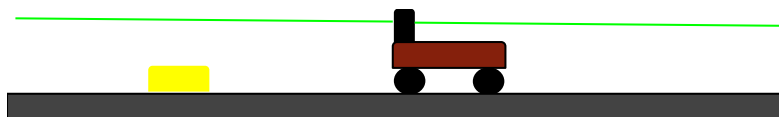
/cmd_vel is the primary channel for movement instructions. It uses the “twist” message type from the “geometry_msgs” package. The content of these messages are two vectors: “linear”, for translational movement in 3D-space, and “angular”, for rotational movement. This topic will evidently be frequently used for robot movement instructions.

The /odom topic is used for odometry, keeping track of the relative position of the robot from its starting point. The message type is “odometry”, and it uses the geometry_msgs types “twist” and “pose” via composition. twist is discussed above, it stores velocity data. Pose contains a “point” data type, storing X Y and Z coordinates, whereas a “quaternion” stores the rotation data (only “yaw” will be relevant in 2D). It is worth noting that the odometry data processed here may become inaccurate due to physical failures such as wheels slipping.

Lastly, the /scan topic is where LiDAR data is passed through. The package “sensor_msgs” provides the message type “LaserScan”, which contains a large amount of data capable of reconstructing a full 2D map of the area visible to the robot. These data include angle separation between scans, time data, minimum and maximum ranges, and a full array of scanned ranges. In order to perform additional actions, such as acquiring the cargo, actuating the lights etc., additional topics are required.

Question 5:

A flaw noticed in the ECED3901 general robot design is its usage of only LiDAR as a means of mapping for navigation. The ECED3901 robot only uses a LiDAR tower at a fixed height, limiting its ability to internally model its surroundings within a singular 2-D plane. This can create navigation & visibility hiccups, as any physical entity not encapsulated within that 2-D plane will *not* be seen by the final LiDAR visualization.



Design Flaw: 2-dimensional Object Detection Shortcomings

A necessary object that must be detected may need to have an additional, alternative, method to its LiDAR system in order for the ECED3901 robot to accurately and reliably complete its mission.